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Exploratory Data Analysis

ACDC member segmentation: evidence before clustering

Data quality, behavioural signals, engineered features and group hypotheses

Group 10

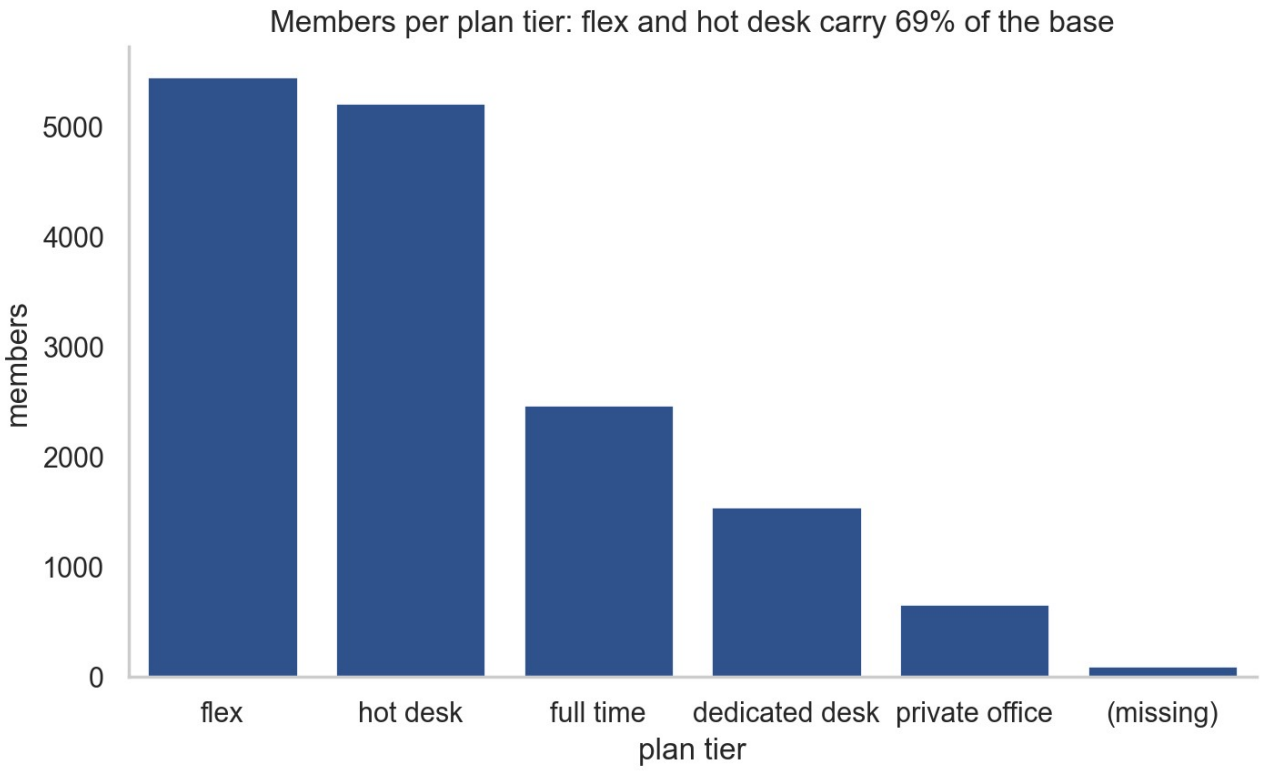
Gustavo Carvalho, Diogo Gaspar, Mohamed Triki

Data Mining, NOVA IMS | 15,450 members analysed

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CRISP-DM: Business and data understanding

ACDC needs member groups it can act on, not a price list



The extract has 15,465 rows. Removing 15 exact duplicates leaves 15,450 members.

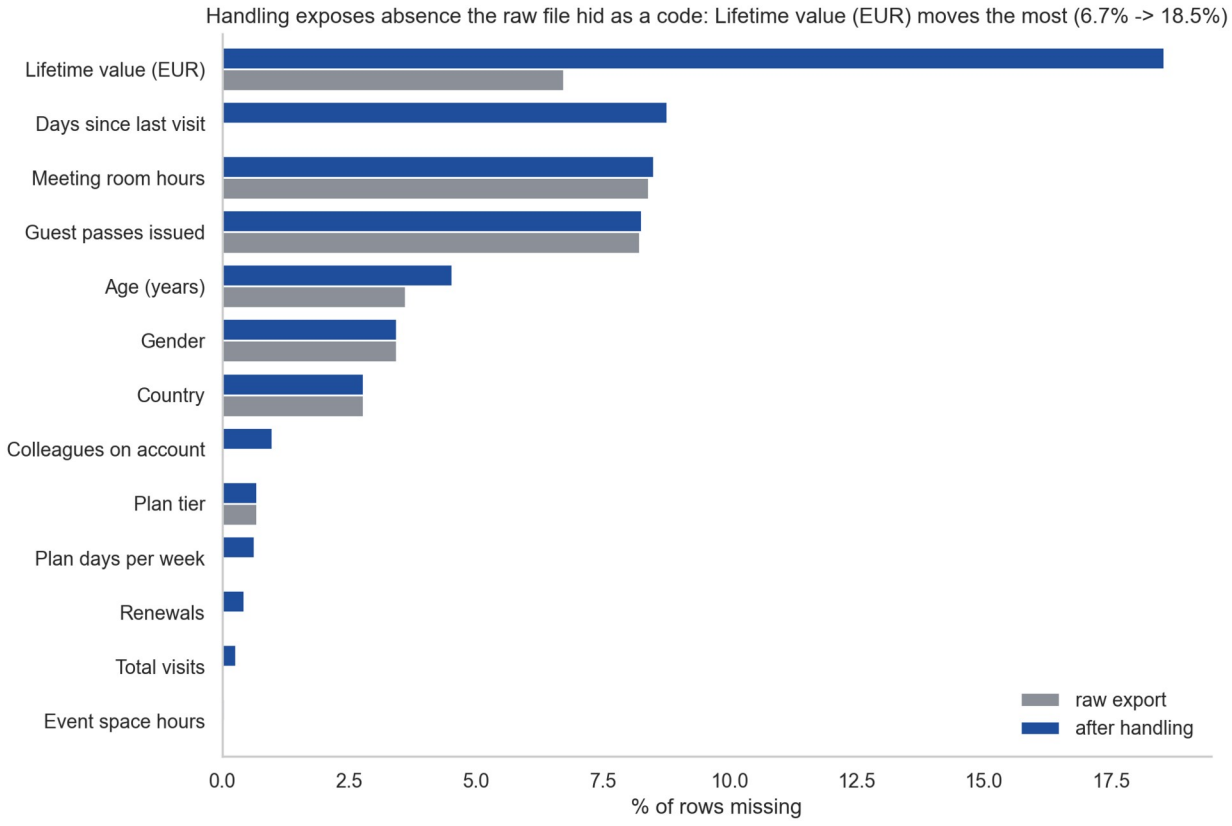
Hot desk and flex hold 69.0% of the base, so plan level alone cannot describe the market.

The data can show value, usage and lifecycle patterns. It cannot prove why a member leaves or assign final segments.

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CRISP-DM: Data understanding

Missing value data is the biggest risk to a reliable grouping



2,864 members, 18.5% of the base, have no usable lifetime value.

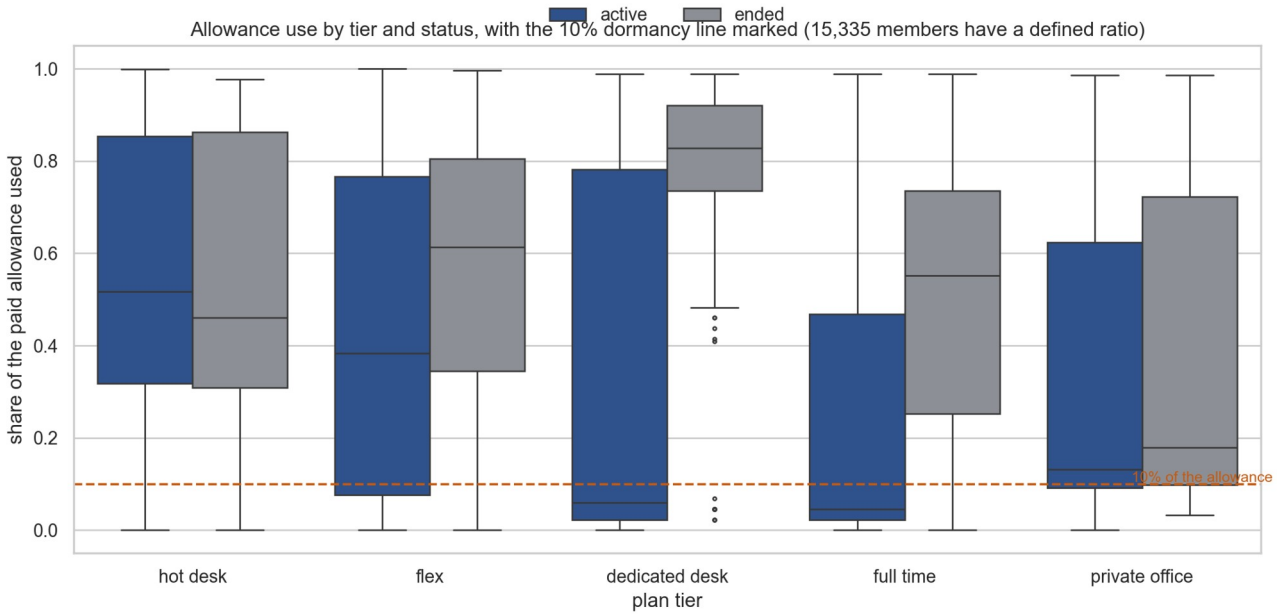
Documented placeholders become missing and keep a flag. Valid extremes are kept.

A missing value leaves a member poorly placed on one axis. A placeholder read as real can create an artificial group.

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CRISP-DM: Data understanding

Usage separates members more clearly than the documented plan ladder



Full time earns a median 320 EUR per month, below dedicated desk at 406 EUR.

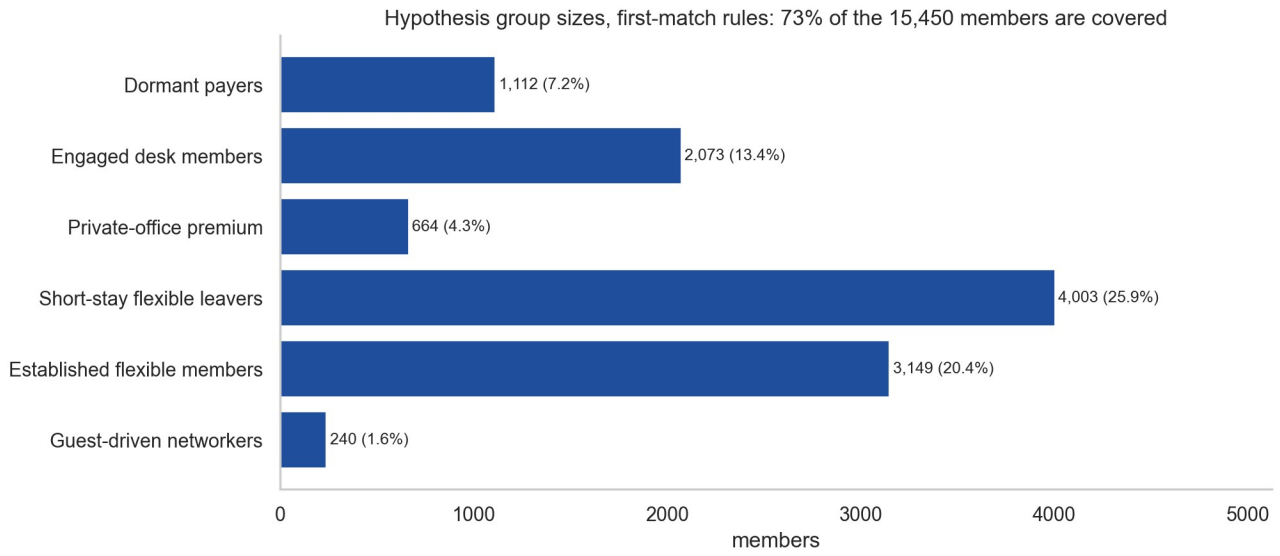
1,112 of 1,804 active committed members use under 10% of their allowance.

Monthly revenue, visits per month, allowance used and recency add behaviour that raw plan fields do not carry.

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CRISP-DM: Customer-group hypotheses

Five revenue levels support a working range of 4 to 6 groups



Six written hypotheses cover 72.8% of members; 27.2% match no rule.

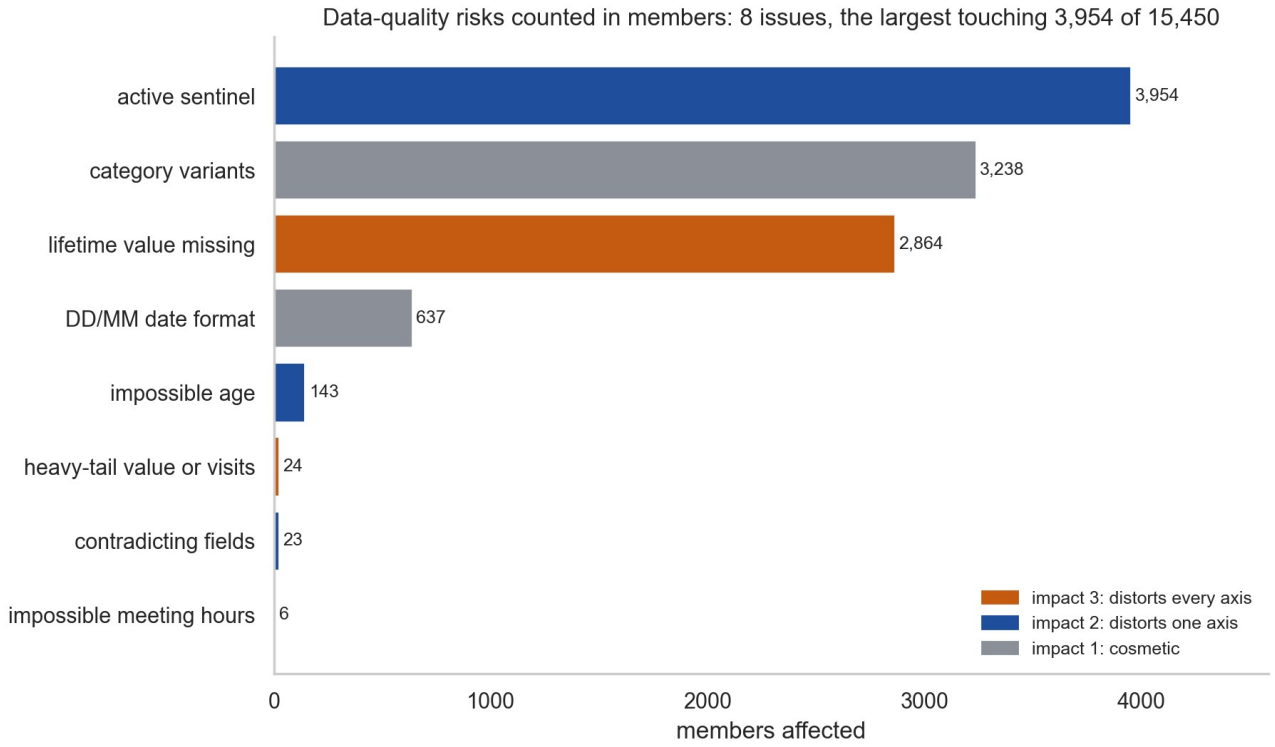
The strongest behavioural differences are visits per month, recency and monthly revenue.

Challenge: 17 suspect lifetime values alone hold 29.5% of recorded revenue.

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CRISP-DM: Communication and next steps

Tell ACDC to target behaviour, then validate the groups with clustering



Plain message: a few members carry most revenue, many cheap members churn, and committed members differ sharply in use.

Next: decide missing-value treatment, scale numerical features, encode categories and test 4 to 6 groups.

Compare a centroid method with a hierarchical method, then check both against the six behavioural hypotheses.